



Fig. 2: Trends in energy requirement and peak load of India over last ten years

a significant and growing requirement for mechanical and electrical engineering diploma holders for installation, commissioning and certification of solar modules and farms."

Commenting on why engineers with any background would be able to take up jobs in solar sector, Shankar says, "It pretty much involves all fields of engineering. However, an engineer with electronics or electrical background would be able to appreciate the system with circuit parameters like current, voltage, power and resistance."

Since it has more to do with power generation, those with power engineering background can also play a great role in doing feed-in systems.

Roles in the solar sector

Solar sector offers many varied roles, including solar energy system installer or designer, sales consultant, installation electrician, engineering technician, photovoltaic design engineer, fabrication technician and installation foreman.

Singh says, "Entry-level positions start from trainee engineers in manufacturing, system and project design, installation, operation and maintenance, procurement, etc."

Adding to the kind of role candidates would perform, Ganesh Shankar says, "Field job may not require great qualification, and a diploma holder may be able to perform as good as an engineer. Product design, on the other hand, would be better suited for engineers with a good knowledge of circuit design."

"Technicians will be required for assembly testing of cells and

with initial exposure to working at site."

For the position of sales and marketing manager, those who have an experience of three to five years in the solar industry are preferred. A sales and marketing manager is required to research and develop marketing strategy, identify business partners like distribution agents, contractors and vendors for supporting sales and delivery, organise service, maintenance and installation, support the local market and prepare budget for the business. He is also accountable for profit and loss.

R&D of solar systems that are tailor-made for Indian conditions could also be a great career opportunity as enough homework is apparently not done in India before the technology is adopted.

Pay package

"Solar power being an industry where domain knowledge and research form an integral part, attracting and retaining the right talent is very important. Therefore pay package here is one of the best in the industry," says Singh. A solar PV design engineer's salary ranges from Rs 325,000 to Rs 520,000 per annum. In the case of a solar power engineer the basic pay starts from Rs 300,000, and for a solar system administrator it varies from Rs 125,000 to Rs 850,000 depending upon his experience.

"At Consul, we advocate that the pay package should not be the sole criteria in deciding on a job. One should also look at the company, its prospects and the growth opportunity it can pro-

vide," says Ramakrishnan.

"Today, the market for engineers is in a very bad shape. The good part is that many engineers are interested in core engineering," says Nayak. "One more issue is employability. Only 30 per cent of the engineering graduates are employable, which has resulted in lower entry-level salaries. Monthly salaries for diploma technicians are in the range of Rs 10,000 to Rs 15,000. Graduate engineers earn Rs 20,000 to Rs 25,000," he adds.

In the near future...

There is a significant emphasis from the government's side on training and creating a pool of people for the job opportunities in the renewable and solar sector. "The Directorate General of Employment and Training (DGET) of Ministry of Labour & Employment has been tasked to incorporate renewable energy in the syllabus of two-year regular ITI courses of seven trades, namely, electrician, electronics, fitter, turner, sheet metal works, plumber and welder," says Ramakrishnan.

Under its human resource development programme, the ministry supports educational institutions in developing the necessary framework for undertaking renewable energy courses at undergraduate, postgraduate and PhD levels.

Major initiatives include fellowship, sponsorship of chairs at IIT Bombay, designation of renewable energy chairs at select institutions, and grant for upgrading library and laboratory facilities. In addition, course curricula and study materials are being developed for postgraduate, undergraduate, diploma and ITI levels.

"Also support is being provided to the state nodal agencies and NGOs for undertaking short-term training programmes besides programmes for capacity building. Short-term training programmes are being organised at block and district levels to attract youth from rural areas to work in installation, operation, maintenance and repair departments," says Ramakrishnan. ●

The author is a tech correspondent at EY Bengaluru